Dr. Kyunghoon HAN

webpage: https://kyunghoon-han.github.io

Computational Physicist · Scientific Computing & Large-Scale Modelling · Applied mathematician

Citizenship: Canadian   Current Residence: Luxembourg   Phone: +352 661 171 771   e-mail: kyunghoon.h@gmail.com

Computational physicist, applied mathematician and machine learning scientist with experience leading research and building quantitative computational workflows across academia, government, and industry. My work has included satellite and remote-sensing data processing at the Canada Centre for Remote Sensing, Natural Resources Canada, reproducible HPC pipelines for simulation and model evaluation in academic research, and leadership as Head of Research at SRuniverse, where I led development of multimodal AI systems across NLP, speech, image, and video. I am interested in bringing this combination of scientific computing, quantitative modelling, and research leadership to real-world problems.

SELECTED EXPERTISE

Category theory-based computation	Theoretical chemical physics, many-body systems
Generative AI & multimodal systems	PyTorch, Python, C/C++, Julia, WebAssembly, Rust, Go
Scientific Computing & Machine Learning	HPC/Slurm, CUDA, Docker, Kubernetes, Jenkins
Large-scale experimentation & reproducibility	Statistical modelling, information theory, optimisation
Model evaluation, ablation & benchmarking	Production-oriented research workflows

EMPLOYMENT

(Post-)doctoral Researcher · University of Luxembourg *2025 — Present (Post-doc), 2021 — 2025 (Doctoral)*

- Developed and benchmarked ML-based force-field workflows, including SO3LR, for molecular simulation, vibrational spectroscopy, and stability analysis; combining quantum-accuracy targets with large-scale computational pipelines (arXiv:2601.09845, under review by *Nature Communications*).
- Built reproducible HPC workflows for large-scale simulation, automated data processing, and model evaluation across cluster environments.
- Applied information-theoretic, geometric (Euclidean and Non-Euclidean; from my thesis), and stochastic methods to model analysis and high-dimensional scientific data; led a partial information decomposition study of electronic observables (arXiv:2603.05672).
- Compared ML-based and physics-based models for accuracy, efficiency, and robustness across diverse molecular workloads (Chapter 11 of arXiv:2605.14974).
- Built and validated computational pipelines for vibrational density-of-states, spectroscopic peak analysis, and force-field benchmarking using ORCA, AMBER, and custom Python tooling.
- Published *TIHI Toolkit* (ACS Omega (2024), **9**(50): 49397–49410) — an open-source peak finder and analyser for spectroscopic data; contributed to five peer-reviewed publications and two preprints.
- Implementation of computer-science friendly description of chemistry via category theory
- *Teaching*: Computational Methods (Master's level, main instructor).

Head of Research / AI Scientist · SRuniverse *2019 — 2021*

- Led research and system integration for multimodal AI pipelines spanning NLP, TTS/STT, image generation, and video generation, contributing to production-facing virtual personalities.
- Built end-to-end pipelines for training, inference, evaluation, and deployment across multiple generative AI components.
- Managed deployment workflows and supporting infrastructure using Python, C/C++, Julia, PyTorch, Docker, Kubernetes, and Jenkins.
- Led a project on development of YouTube personality *RuiCovery* (sample link)
- Led iterative model experimentation focused on robustness, integration quality, and readiness for deployment.

Research Assistant · Canada Centre for Remote Sensing, Natural Resources Canada *2010 — 2011*

- Built signal-processing pipelines for RADARSAT-2 satellite data acquisition; implemented relativistic and Doppler image corrections using Canadian Digital Elevation Models.
- Maintained satellite transponders, performed orbit prediction, and modernised legacy C radio signal-processing code (C for Windows 3.2 → modern C).
- Satellite orbit interpolation algorithm from sparse velocity-position data.

Earlier Experience

Researcher, Hankook Life Science Institute, Seoul (2017–2018) · Teacher, Steven Academy, Seoul (2016, 2018) · English Translator, Hankook Tire, Daejeon (2015)

EDUCATION

Doctor of Physics
University of Luxembourg · 2021—2025
Theoretical Physics
Thesis: Analysis, Calculation and Understanding of Molecular Vibration

Bachelor of Mathematics (Honours)
University of Waterloo · 2008—2014
Pure Mathematics & Mathematical Physics (Double Major)
Thesis: Osmotic compaction of bacterial chromosomes

Master of Science
Université de Tours · 2016—2017
Nonlinear Models in Theoretical Physics
Thesis: Simulation of vortex behaviour in the inner-crust of a neutron star

FIVE RECENT WORKS

1. Han, K. (2026). Categorification of Chemical Reactions: a bottom-up tower from stoichiometry to quantum structure. *arXiv preprint arXiv:2605.14974*.

2. Han, K. & Gallegos, M. (2026) Partial information decomposition of electronic observables along a reaction coordinate. arXiv preprint arXiv:2603.05672

3. Han, K., Tkatchenko, A. & Berryman, J. T. (2026) Symplectic and Thermodynamically Consistent Molecular Dynamics in the Frequency Domain—under review by *Physics Review Letters*

4. Suárez-Dou S., Gallegos M., Han K. et al. (2026) Stability and vibrations of proteins in vacuum and water: bridging quantum accuracy and force-field efficiency. arXiv:2601.09845—under review by *Nature Communications*

5. Dey, S., Field, E. H., Wang, Y., Han, K., ... & Bera, S. (2025). Fluorination Induced Inversion of Helicity and Self-Assembly Into Cross-α Like Piezoelectric Amyloids by Minimalistic Designer Peptide. *Small*, *21*(18), 2500288.

Full publication list available on Google Scholar.

ACADEMIC TALKS

Chemical Reaction Networks in Hawai'i 2026 Hawai'i, May 19th, 2026 (link)	Open Seminar on Korean Scientists and Engineers Association in Luxembourg (link) Lux., Apr. 18th, 2026 **INVITED TALK**	German Physical Society (DPG) Spring Meeting Dresden, Mar. 10th 2026 (link)	IEEE Aerospace and Electronics Systems Society, Ottawa Chapter (invitation letter can be provided) Ottawa, Nov. 26th,2025 **INVITED TALK**
Categorification of Chemical Reactions: from stoichiometry to quantum structures	*Partial information decomposition of electronic observables along reaction pathways*	*On information theoretic re-definition of chemical bonds using partial information decomposition*	*On quantum radars and partial information decomposition*

LANGUAGES

English — fluent (native)
Korean — fluent (native)
French — intermediate
Japanese — intermediate
German — B1

References and full CV available on request.